

Optimal control for daily operation planning of a water distribution system with pumped storage

Mario Vašak
University of Zagreb, Faculty of
Electrical Engineering and Computing,
Laboratory for Renewable Energy Systems
Zagreb, Croatia
mario.vasak@fer.hr

Hrvoje Novak
University of Zagreb, Faculty of
Electrical Engineering and Computing,
Laboratory for Renewable Energy Systems
Zagreb, Croatia
hrvoje.novak@fer.hr

Abstract — Approaching a carbon-free power system mandates its demand-side to become flexible in shaping the consumption profiles. The demand re-shape should be based on economical grounds, i.e. the flexibility should reduce the operational costs of the participating consumer. Besides industrial facilities or buildings where the demand flexibility should not infringe the production or comfort, also different infrastructures can be considered for flexibilization of their consumption profiles. Also in this case the overall operational costs of the infrastructure should be reduced while basic performance constraints must be respected. In this paper we focus on optimization of daily operation of a water distribution system with a pumped storage. The cumulative cost of electricity consumption and water leakages in the system is minimized while the pressures in all the end-points of the network are kept above the minimum prescribed levels.

Keywords — water distribution system, pumped storage, optimal control, daily operation planning, electricity cost, pressure, water leakage

I. INTRODUCTION

Water distribution systems (WDSs) are complex dynamical systems consisting of pumps, storages, pipes and valves. The main goal in their operation is to satisfy water demands in different customer connection points while ensuring all these end-points are at a sufficiently high pressure level. However, keeping pressures at significantly higher levels than minimum required substantially increases water losses on all places where leaks occur [1].

Water distribution systems are in majority of cases managed by relying on experience of operators who decide on operation of pumps and valves in the system. The operators usually operate the system in a rather conservative way – with a set of control actions that ensure proper pressure levels in different end-points of the system with significant margins. This contributes to the water distribution system operation costs in terms of both energy and water losses, and is thus a motivator for introduction of optimal control strategies in it.

Optimal control strategies in this particular application should weigh electricity consumption costs and water leakages in the water distribution system while keeping all pressures above the minimum thresholds. However, there is even more to it in the era of rising needs for demand response – it is the possibility to use the optimal control algorithm to plan the system operation in demand response scheme.

The relations for modelling flows and pressures in various WDS points are nonlinear and this makes the potential optimal control schemes applied to water distribution system significantly complex. In general they can be divided into three parts [2] – (i) hydraulic modelling for the optimal control problem, (ii) prediction of water demand in different points of the network, and (iii) the formulation/solution of the optimal problem itself.

Prediction of water demand in different network points usually relies on artificial intelligence techniques like neural networks for historical data processing and creating the prediction model based on them. Numerous researches were conducted in that area and it becomes more or less a commercially ready technology.

In this paper we foremost focus on the formulation and solution of the optimal control problem for a case study of the pumped storage. We employ sequential linear programming (SLP) technique to employ sequential linearization of a rather complex hydraulic model for pressure-flow dependence in the piping of the WDS. The linearization is employed in such a way that maximum possible accuracy compared to the starting nonlinear model is maintained: a separate linear model is determined at each time step of the prediction horizon with ensured continuity of the states of the models between samples while the energy consumption and water leakage are determined by parametrically integrating the response of the linearized models. Also efficient simulation of the behaviour of the nonlinear model along the prediction horizon is employed in order to assess the linearization points for consecutive application of SLP.

The optimization problem is formulated over a 24-hours long prediction period, and periodicity constraint for the system state – water level in the tank – is enforced, with the aim to make the optimized behaviour of the system repeatable on a daily scale. The optimization problem decides on the pumping schedule, but also simultaneously on the initial/end height of the water in the tank.

The paper is organized as follows. Section II introduces the considered water distribution system case study with pumped storage and its corresponding modelling. The formulation of the optimization problem for SLP is given in Section III. Section IV outlines the results obtained for the optimized behaviour of the water distribution system on a daily scale. Section V concludes the paper.

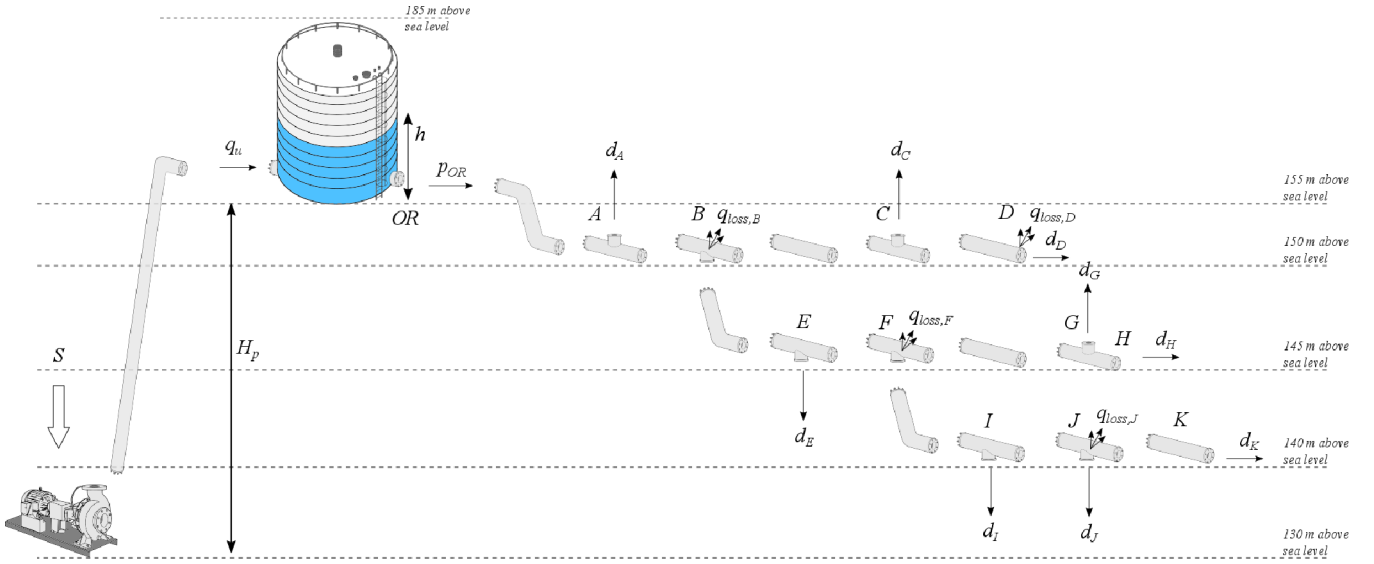


Fig. 1. Configuration of the considered WDS.

II. WATER DISTRIBUTION SYSTEM CASE STUDY

A. System configuration

Configuration of the WDS for the case study is shown in Fig. 1. It is a system with a single storage, a pump and the piping branched from the tank outlet. The parameters of the case study system are provided in TABLE I. and particularly the parameters related to piping are provided in TABLE II. Pipes diameters are determined based on presumed maximum water demands such that the linear velocity of water in the pipes does not exceed 3 m/s and that the pressure drop along pipes allows to attain minimum allowed pressure in all end-points with maximum pressure nowhere exceeding the amount $5P_0$ [3].

TABLE I. PARAMETERS USED IN THE WDS CASE STUDY

Parameter	Description	Value
ν	Kinematic coefficient of water viscosity (at 15°C)	$1.15 \cdot 10^{-6} \text{ m}^2/\text{s}$
ρ	Density of water	1000 kg/m^3
g	Gravity acceleration	9.81 m/s^2
A_{leak}	Area of leak surfaces on pipes	10^{-5} m^2
P_0	Atmospheric pressure	101325 Pa
A	Area of the tank cross section surface	10 m^2
$Q_{u,\text{max}}$	Maximum flow of the pump	$2.78 \cdot 10^{-2} \text{ m}^3/\text{s}$ (100 m ³ /h)
P_{min}	Minimum allowed pressure at end-points	$2P_0=202650 \text{ Pa}$
H_{max}	Tank height from bottom to top	30 m
η	Pump efficiency	0.85
H_p	Elevation of the tank bottom above the pump	25 m
D_{max}	Maximum demand at different piping supply points	$2.78 \cdot 10^{-3} \text{ m}^3/\text{s}$ (10 m ³ /h)

The WDS serves the demands in the following 9 points: A, C, D, E, G, H, I, J, K. These demands are referred to as $d_A, d_C, d_D, d_E, d_G, d_H, d_I, d_J, d_K$, respectively. Flows through different pipes are denoted with endpoints of particular pipes

TABLE II. WATER DISTRIBUTION SYSTEM PIPING PARAMETERS

Parameter	Description	Value
d_{OR_A}	Pipe diameter from output of the reservoir (OR) to point A	0.125 m
d_{A_B}	Pipe diameter from A to B	0.125 m
d_{B_C}	Pipe diameter from B to C	0.1 m
d_{C_D}	Pipe diameter from C to D	0.1 m
d_{B_E}	Pipe diameter from B to E	0.125 m
d_{E_F}	Pipe diameter from E to F	0.125 m
d_{F_G}	Pipe diameter from F to G	0.1 m
d_{G_H}	Pipe diameter from G to H	0.065 m
d_{F_I}	Pipe diameter from F to I	0.1 m
d_{I_J}	Pipe diameter from I to J	0.1 m
d_{J_K}	Pipe diameter from J to K	0.065 m
l_{OR_A}	Pipe length from OR to A	100 m
l_{A_B}	Pipe length from A to B	100 m
l_{B_C}	Pipe length from B to C	300 m
l_{C_D}	Pipe length from C to D	200 m
l_{B_E}	Pipe length from B to E	300 m
l_{E_F}	Pipe length from E to F	300 m
l_{F_G}	Pipe length from F to G	300 m
l_{G_H}	Pipe length from G to H	200 m
l_{F_I}	Pipe length from F to I	300 m
l_{I_J}	Pipe length from I to J	200 m
l_{J_K}	Pipe length from J to K	200 m
Δh_{OR_A}	Height difference between OR and A	5 m
Δh_{A_B}	Height difference between A and B	0 m
Δh_{B_C}	Height difference between B and C	0 m
Δh_{C_D}	Height difference between C and D	0 m
Δh_{B_E}	Height difference between B and E	5 m
Δh_{E_F}	Height difference between E and F	0 m
Δh_{F_G}	Height difference between F and G	0 m
Δh_{G_H}	Height difference between G and H	0 m
Δh_{F_I}	Height difference between F and I	5 m
Δh_{I_J}	Height difference between I and J	0 m
Δh_{J_K}	Height difference between J and K	0 m
ϵ	Pipe roughness	0 m

in subscript, i.e. they are denoted as follows: $q_{\text{OR}_A}, q_{A_B}, q_{B_C}, q_{C_D}, q_{B_E}, q_{E_F}, q_{F_G}, q_{G_H}, q_{F_I}, q_{I_J}, q_{J_K}$. The leaks are

assumed to occur in points B, D, F and J and are denoted with $q_{\text{loss,B}}$, $q_{\text{loss,D}}$, $q_{\text{loss,F}}$ and $q_{\text{loss,J}}$, respectively.

B. Water distribution system model

The WDS dynamic model resides on the water mass balance equation in the tank:

$$\frac{dh}{dt} = \frac{1}{A}(q_u - q_{\text{OR,A}}), \quad (1)$$

where t [s] denotes time, h [m] is the height of water in the tank and q_u [m³/s] is the input flow from the pump into the tank. Relation between pressures on end-points X and Y of a certain pipe with fluid flowing from point X to point Y is [4]

$$p_Y = p_X + \rho g \Delta h_{X,Y} - \alpha \frac{l_{X,Y}}{d_{X,Y}^{4.75}} \cdot q_{X,Y}^{1.75}, \quad (2)$$

where $\Delta h_{X,Y}$ is the difference in elevation between points X and Y and α [Pa·s^{1.75}/m^{0.5}] is calculated as

$$\alpha = 0.241434 \rho^4 \sqrt{V}. \quad (3)$$

We assume that the tank is open such that the water top in the tank is at the pressure P_0 . Thus the pressure at point OR (output of the tank) is

$$p_{\text{OR}} = P_0 + \rho g h. \quad (4)$$

The leakage flow at certain point X is modelled as [1]

$$q_{\text{loss,X}} = A_{\text{leak}} \sqrt{\frac{2}{\rho} (p_X - P_0)}. \quad (5)$$

The flow through a certain pipe is equal to sum of all demands and leaks downstream the pipe, e.g.

$$\begin{aligned} q_{\text{OR,A}} &= d_A + d_C + d_D + d_E + d_G + d_H + d_I + d_J + \\ &\quad + d_K + q_{\text{loss,B}} + q_{\text{loss,D}} + q_{\text{loss,F}} + q_{\text{loss,J}}, \\ q_{\text{E,F}} &= d_G + d_H + d_I + d_J + d_K + q_{\text{loss,F}} + q_{\text{loss,J}}. \end{aligned} \quad (6)$$

Assuming that the pressure in water entering the pump is P_0 , the electrical power of the pump S [W] is calculated as

$$S = \frac{1}{\eta} \rho g (H_p + h) q_u. \quad (7)$$

The pump is driven via variable frequency drive such that the flow can be continuously set from 0 to $Q_{u,\text{max}}$. Model of the pump is simplified and it is assumed that the pump system can provide the maximum flow for any water height in the tank up to H_{max} .

III. OPTIMAL CONTROL PROBLEM FORMULATION AND SOLUTION

A. Nonlinear optimal control problem

The operation cost of the WDS in a certain period of time corresponds to the sum of costs for the spent electrical energy on the pump and for the volume of lost water through leaks during that period. The goal in operation of the WDS is to satisfy all demands with sufficient pressure in all end-points, while keeping the operation cost at minimum. The methodology of optimal control is employed to compute the optimal profile of the input flow q_u for a certain prediction

horizon as well as to decide on the optimal starting water height in the tank.

The computation of optimal controls is performed on a control computer which considers states of the system in a finite number of time instants influenced with a finite number of control inputs and requires a finite time for re-computation of the optimal control sequence in a receding horizon fashion [5] when switching from planning to operation. The system is thus discretized with sampling time T , and the input flow q_u is maintained constant within each time step $[kT, (k+1)T)$,

$$q_u(t) = q_{u,k} \quad \forall t \in I_k = [kT, (k+1)T), \quad k \in Z, \quad (8)$$

where Z is the set of integers. The cost function for the control problem is defined as follows:

$$J = \sum_{k=0}^{N-1} p_{e,k} E_k + p_w W_k, \quad (9)$$

where $p_{e,k}$ [EUR/kWh] is the price of electricity in time interval I_k , p_w [EUR/m³] is the price of water, NT is the prediction horizon, E_k [kWh] is the electrical energy spent on the pump in time interval I_k , i.e.

$$E_k = \frac{1}{1000 \cdot 3600} \int_{kT}^{(k+1)T} S(t) dt, \quad (10)$$

while W_k [m³] is the volume of lost water in I_k , i.e.

$$W_k = \int_{kT}^{(k+1)T} (q_{\text{loss,B}}(t) + q_{\text{loss,D}}(t) + q_{\text{loss,F}}(t) + q_{\text{loss,J}}(t)) dt. \quad (11)$$

The lowest pressures in the WDS occur at end-points as long as there is no up-hill water path after the tank output. Since there are none such paths in the WDS considered, it is enough to pose pressure constraints at the end-points D, H and K. Although pressure constraints will be only posed at the discrete-time instants kT , the aperiodic behaviour of the system under constant input flow q_u within the time interval will assure that the constraints are maintained throughout the prediction horizon. The posed pressure constraints are:

$$\begin{aligned} p_{D,k} &\geq P_{\min}, \quad k=0, \dots, N, \\ p_{H,k} &\geq P_{\min}, \quad k=0, \dots, N, \\ p_{K,k} &\geq P_{\min}, \quad k=0, \dots, N, \end{aligned} \quad (12)$$

where $p_{X,k} = p_X(kT)$. Similarly the constraints are posed also for the water heights in the tank along the prediction horizon:

$$0 \leq h_k \leq H_{\max}, \quad k=0, \dots, N, \quad (13)$$

where $h_k = h(kT)$. Constraints are also posed for the input flow from the pump:

$$0 \leq q_{u,k} \leq Q_{u,\text{max}}, \quad k=0, 1, \dots, N-1. \quad (14)$$

Finally, the periodicity constraint is imposed such that the water height at the end of the prediction horizon is equal to the height at the beginning:

$$h_0 = h_N. \quad (15)$$

The nonlinear optimization problem can be now posed as follows:

$$\begin{aligned} \min_{h_0, q_{u,0}, q_{u,1}, \dots, q_{u,N-1}} J \\ \text{subject to (12)-(15).} \end{aligned} \quad (16)$$

Note that the optimization problem (16) addresses not just the pumping sequence, but also the optimal starting height of the water in the tank since it is focussed on WDS daily operation planning. The sampling time T is selected to be $T=15$ min in order to correspond to the presumed period of change of the electricity prices, and the prediction horizon is selected to be 1 day long, i.e. $N=96$. In this way one can for predicted demand sequences in the WDS supply points determine the optimal behaviour of the WDS. Here focus is not on techniques for water demand prediction and it is assumed that these prediction sequences can be obtained based on historical data for different day types and predicted weather conditions, see e.g. [6].

Due to the nonlinearity of the WDS model, the problem (16) cannot be easily solved. The WDS model thus needs to be linearized and then the problem (16) replaced with an approximate problem based on linearization performed. With such problem it will be possible to decide on the direction of change of the optimization variables and finally to reach the optimum through iterations.

B. Linearization of the water distribution system model

For a fixed water height H_0 , input flow Q_{u0} and demands $D_{A0}, D_{C0}, D_{D0}, D_{E0}, D_{G0}, D_{H0}, D_{I0}, D_{J0}, D_{K0}$ the linearization of the model is performed in order to approximate the behaviour of pressures, flows and leaks in the WDS for small changes in water height or input flow.

First all pressures, flows and leaks need to be determined for given height and demands. This is done by numerically solving the system of equations (2) with insertions from (3) to (6) which gives 11 equations for pressures in different points of the WDS with 11 unknowns being these pressures, i.e. $P_{A0}, P_{B0}, P_{C0}, P_{D0}, P_{E0}, P_{F0}, P_{G0}, P_{H0}, P_{I0}, P_{J0}, P_{K0}$. After they are determined, the losses and flows are determined by applying relations (5) and (6), consecutively. Recall that (6) contains only two exemplary relations, while actually there are 11 different relations for flows. The determined losses are: $Q_{\text{loss},B,0}, Q_{\text{loss},D,0}, Q_{\text{loss},F,0}, Q_{\text{loss},J,0}$. The determined flows are: $Q_{\text{OR},A,0}, Q_{A,B,0}, Q_{B,C,0}, Q_{C,D,0}, Q_{B,E,0}, Q_{E,F,0}, Q_{F,G,0}, Q_{G,H,0}, Q_{F,I,0}, Q_{I,J,0}, Q_{J,K,0}$. The procedure for determining the pressures, losses and flows in the operating point (OP) we will denote shortly with:

$$[P_{X,0}, Q_{\text{loss},X,0}, Q_{X,Y,0}] = \text{OP}(H_0, D_{X,0}). \quad (17)$$

The linearization of the considered WDS is performed further as follows. For the water height h , each pressure p_X , loss $q_{\text{loss},X}$ and flow $q_{X,Y}$ one can write the following:

$$\begin{aligned} h &= H_0 + \Delta h \\ p_X &= P_{X,0} + \Delta p_X, \\ q_{X,Y} &= Q_{X,Y,0} + \Delta q_{X,Y}, \\ q_{\text{loss},X} &= Q_{\text{loss},X,0} + \Delta q_{\text{loss},X}, \end{aligned} \quad (18)$$

where $\Delta h, \Delta p_X, \Delta q_{X,Y}$ and $\Delta q_{\text{loss},X}$ are small changes around the operating point. By inserting (18) into (2), (4), (5) and (6) and by approximating all the nonlinear mappings with a first-order Taylor polynomial, the following linear relations are obtained for the small changes:

$$\begin{aligned} \Delta p_X &= K_X \Delta h, \\ \Delta q_{\text{loss},X} &= K_{\text{loss},X} \Delta h, \\ \Delta q_{\text{OR},A} &= K_h \Delta h, \end{aligned} \quad (19)$$

where $K_X, K_{\text{loss},X}$ and K_h are the computed linearization coefficients.

For a fixed sequence of flows $q_{u,k}=Q_{u,k,0}$ and predicted demands $d_{X,k}$ along the prediction horizon, i.e. $k=0, \dots, N-1$, and for initial water height in the tank $H_{0,0}$, first a sequence of heights $H_{k,0}$ is obtained by performing simulation of the nonlinear model via some numerical integration procedure. For each of the operating points defined with $H_{k,0}$ and $D_{X,k}$ the pressures, flows and losses are computed by following (17). After that linearization is performed with introduced changes of input flow and height

$$\begin{aligned} q_{u,k} &= Q_{u,k,0} + \Delta q_{u,k}, \\ h_k &= H_{k,0} + \Delta h_k, \end{aligned} \quad (20)$$

such that finally linearization coefficients $K_{X,k}, K_{\text{loss},X,k}$ and $K_{h,k}$ can be computed. With computed coefficients $K_{h,k}$ also a sequence of linear dynamical WDS models along the prediction horizon adhering to the following difference equation can be found:

$$\frac{dh(t)}{dt} = \frac{1}{A} (\Delta q_{u,k} - K_{h,k} \Delta h(t)), \quad t \in I_k. \quad (21)$$

For each time interval $I_k, k=0, \dots, N-1$, the following relation for water height h is obtained:

$$\begin{aligned} h(t) &= H_{k,0} + \Delta h_k e^{-\frac{K_{h,k}}{A}(t-kT)} + \\ &+ \left(\frac{1}{K_{h,k}} \Delta q_{u,k} + \beta_k \right) \left(1 - e^{-\frac{K_{h,k}}{A}(t-kT)} \right), \quad t \in I_k, \end{aligned} \quad (22)$$

where $\beta_k = \frac{H_{k+1} - H_k}{1 - e^{-\frac{K_{h,k}}{A}T}}$. The following discrete relation holds for the small changes of height at sampling instants:

$$\begin{aligned} \Delta h_{k+1} &= \Phi_k \Delta h_k + \Gamma_k \Delta q_{u,k}, \\ \Phi_k &= e^{-\frac{K_{h,k}}{A}T}, \quad \Gamma_k = \frac{1}{K_{h,k}} \left(1 - e^{-\frac{K_{h,k}}{A}T} \right). \end{aligned} \quad (23)$$

By introducing

$$\begin{aligned} \Delta \mathbf{h} &= [\Delta h_0 \quad \Delta h_1 \quad \dots \quad \Delta h_N]^T, \\ \Delta \mathbf{q}_u &= [\Delta q_{u,0} \quad \Delta q_{u,1} \quad \dots \quad \Delta q_{u,N-1}]^T, \end{aligned} \quad (24)$$

the following can be written based on (23):

$$\Delta \mathbf{h} = \mathbf{A} \cdot \Delta \mathbf{h}_0 + \mathbf{B} \cdot \Delta \mathbf{q}_u, \quad (25)$$

$$\mathbf{A} = \begin{bmatrix} 1 \\ \Phi_0 \\ \Phi_0 \Phi_1 \\ \vdots \\ \prod_{i=0}^{N-1} \Phi_i \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ \Gamma_0 & 0 & 0 & \dots & 0 \\ \Phi_1 \Gamma_0 & \Gamma_1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \left(\prod_{i=1}^{N-1} \Phi_i \right) \Gamma_0 & \left(\prod_{i=2}^{N-1} \Phi_i \right) \Gamma_1 & \left(\prod_{i=3}^{N-1} \Phi_i \right) \Gamma_2 & \dots & \Gamma_{N-1} \end{bmatrix}.$$

By employing (22) and (25) in (7), (10) and (11) one can now also obtain the expressions for E_k and W_k needed for the

criterion of the cost function of the optimal control problem, now expressed through Δh_0 and $\Delta \mathbf{q}_u$. These are as follows:

$$E_k = \frac{1}{2} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix} \mathbf{A}_{E,k} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T + \mathbf{B}_{E,k} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T + c_{E,k}, \quad (26)$$

$$W_k = \mathbf{B}_{\text{loss},k} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T + c_{\text{loss},k},$$

where $\mathbf{A}_{E,k} \in R^{(N+1) \times (N+1)}$, $\mathbf{B}_{E,k} \in R^{1 \times (N+1)}$, $c_{E,k} \in R$, $\mathbf{B}_{\text{loss},k} \in R^{1 \times (N+1)}$, $c_{\text{loss},k} \in R$, $k = 0, 1, \dots, N-1$, are properly constructed. The overall cost J for the linearized WDS can now be expressed as

$$J = \frac{1}{2} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix} \mathbf{H} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T + \mathbf{F} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T + c, \quad (27)$$

$$\text{where } \mathbf{H} = \sum_{k=0}^{N-1} p_{e,k} \mathbf{A}_{E,k}, \quad \mathbf{F} = \sum_{k=0}^{N-1} p_{e,k} \mathbf{B}_{E,k} + p_w \mathbf{B}_{\text{loss},k},$$

$$c = \sum_{k=0}^{N-1} p_{e,k} c_{E,k} + p_w c_{\text{loss},k}.$$

Constraints for the problem (12)-(15) can now also be expressed by employing (25), e.g.

$$\begin{aligned} p_{H,k} &= P_{H,0,k} + \Delta p_{H,k} = P_{H,0,k} + K_{H,k} \Delta h_k = \\ &= P_{H,0,k} + K_{H,k} \left(\mathbf{A}_{(k+1,:)} \cdot \Delta h_0 + \mathbf{B}_{(k+1,:)} \cdot \Delta \mathbf{q}_u \right) \geq P_{\min}, \end{aligned} \quad (28)$$

where $\mathbf{M}_{(i,:)}$ denotes the i th row of matrix $\mathbf{M}$. Besides constraints (12)-(15), also additional constraints need to be introduced to keep the assumptions of the linearization fulfilled, i.e. to keep Δh_0 and $\Delta \mathbf{q}_u$ sufficiently small. Finally, all constraints stacked together can for the linearized WDS be expressed as

$$\begin{aligned} \mathbf{A}_{\text{ineq}} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T &\leq \mathbf{b}_{\text{ineq}}, \\ \mathbf{A}_{\text{eq}} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T &= \mathbf{b}_{\text{eq}}. \end{aligned} \quad (29)$$

Relations (27) and (29) define a mathematical programme. Usually $\mathbf{H}$ due to the change of the dynamics along the prediction horizon is not positive semidefinite which then requires further to approximate J for small changes Δh_0 and $\Delta \mathbf{q}_u$ with

$$J = \mathbf{F} \begin{bmatrix} \Delta h_0 & \Delta \mathbf{q}_u^T \end{bmatrix}^T + c, \quad (30)$$

finally coming to a Linear Programme (LP) formulation of the WDS perturbation problem with (29) and (30).

Once the LP (29)-(30) is solved, the optimal solution denoted with $\begin{bmatrix} \Delta h_0^* & \Delta \mathbf{q}_u^{*T} \end{bmatrix}^T$ is added to the input flow sequence and the initial height to get their values for the new iteration of optimization:

$$\begin{bmatrix} H_{0,0} \\ \mathbf{Q}_{u,0} \end{bmatrix}_{\text{iter}+1} = \begin{bmatrix} H_{0,0} \\ \mathbf{Q}_{u,0} \end{bmatrix}_{\text{iter}} + \begin{bmatrix} \Delta h_0^* & \Delta \mathbf{q}_u^{*T} \end{bmatrix}^T. \quad (31)$$

The iterative procedure of optimization involving Sequential Linear Programming (SLP) is stopped if the criterion J is decreased for less than the tolerance value or the norm of the optimal solution $\begin{bmatrix} \Delta h_0^* & \Delta \mathbf{q}_u^{*T} \end{bmatrix}^T$ is close to zero.

IV. CASE STUDY RESULTS AND ANALYSIS

The derived optimal control methodology is applied to the WDS case study. The predicted demand profiles for the considered 9 supply points are provided in Fig. 2.

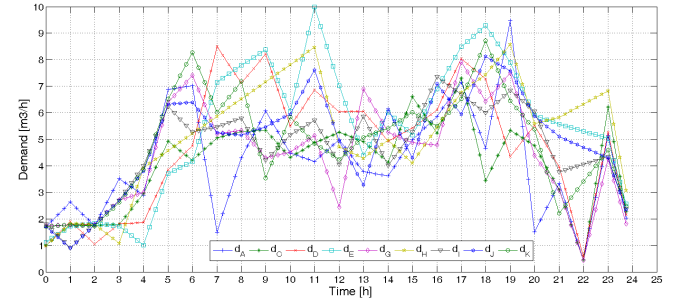


Fig. 2. Predicted demands for the 9 supply points of the considered WDS.

The electricity price profile is shown in Fig. 3. Water price is set at 2 EUR/m³ throughout the day.

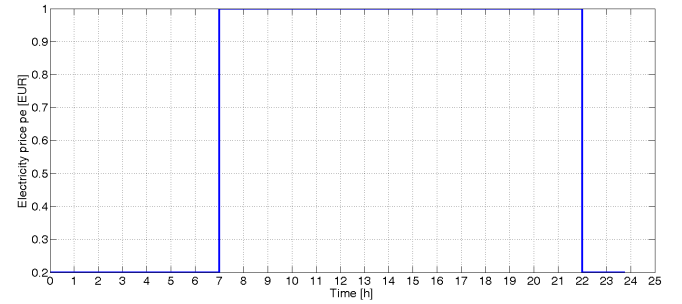


Fig. 3. Electricity price profile.

The optimization of the WDS behaviour is regarding $(\mathbf{Q}_{u,0})_{\text{iter}=1}$ started from a random input flow profile uniformly sampled between 0 and $Q_{u,\max}$, and it is shown in Fig. 4 with the thin line. The initial height $(H_{0,0})_{\text{iter}=1}$ was set at 10 m.

The finally obtained optimal input flow profile after the application of the SLP procedure described in Section III is shown in Fig. 4 with the thick line.

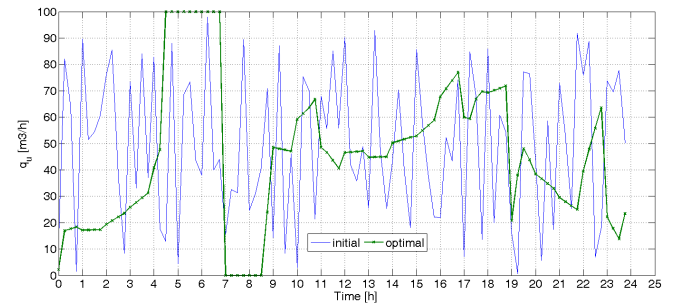


Fig. 4. The initial and computed optimal input flow q_u .

The behaviour of the water height profile h through iterations is shown in Fig. 5. The height profile related to the initial random flow $(\mathbf{Q}_{u,0})_{\text{iter}=1}$ is denoted in the figure and also the final optimal water height profile. One may notice that the initial height has lowered from 10 m to somewhat less than 6 m at the optimum. The optimization procedure also takes care that the initial height corresponds to the final height which makes it possible to use this behaviour repeatedly, i.e. the optimization does not use the initial water height for a single-day economics. One may see the shaped behaviour of the WDS which exploits the variability of

electricity prices to fill the tank during the period of low electricity price. The tank is however not filled to the full ($H_{\max}=30$ m) since in that case the cost for lost water would prevail over the gains in electricity cost.

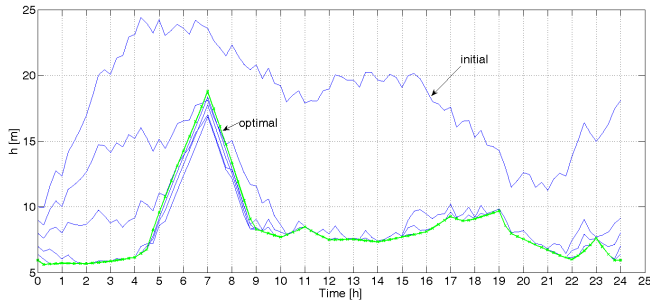


Fig. 5. The initial profile of height h , h profiles through iterations and the optimal h profile.

The behaviour of pressures in end-points of the considered WDS for the optimal solution is shown in Fig. 6. One may notice that the pressure constraints for ensuring quality supply are fulfilled all the time.

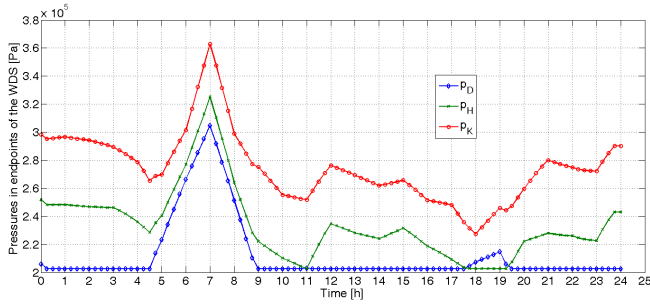


Fig. 6. Optimal profiles of pressure in WDS end-points.

The electricity consumption profile for the optimal behaviour of the WDS is shown in Fig. 7.

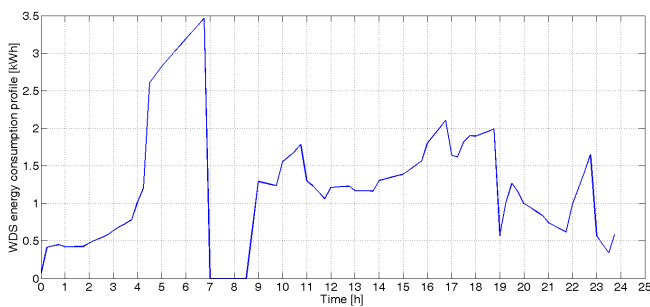


Fig. 7. Optimal electrical energy consumption. The points in the graph at time kT correspond to E_k from (10), i.e. represent 15-minute energy consumption from kT to $(k+1)T$.

Finally Fig. 8 shows the change of the total cost of WDS operation within the 24-hours period through iterations, starting from the initial random flow and 10 m initial height to the optimized behaviour. The maximum allowed changes in h_0 and $q_{u,k}$ between iterations were set at 1 m and $0.2Q_{u,\max}$, respectively. The optimization procedure ended within 10 iterations due to a negligible change of the

optimizer or cost at the final iteration according to the stopping criterion – the threshold was set to 10^{-8} for either the norm of the optimizer change or the cost change.

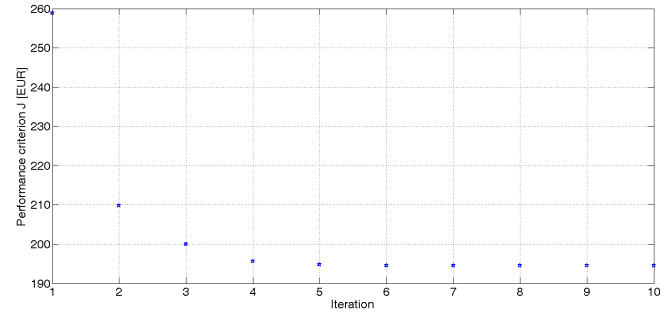


Fig. 8. Total cost of daily operation of the WDS through iterations of the optimization procedure.

V. CONCLUSION

The paper outlines the optimal control methodology for a water distribution system with pumped storage. Nonlinear model of the water distribution system is derived and the methodology for optimal planning of the 24-hours ahead operation of the WDS is presented which resides on linearization and a sequential linear program. The procedure is scalable and applicable for exploitation of the water distribution system infrastructure in demand response and facilitation of renewable energy integration in the energy system.

ACKNOWLEDGMENT

This work was supported in part by the Danube Transnational Programme through the project Smart Building – Smart Grid – Smart City (3Smart, <http://www.interreg-danube.eu/3smart>) under Grant DTP1-502-3.2-3Smart, and in part by the European Regional Development Fund under Grant KK.01.1.1.01.0009 (DATACROSS).

REFERENCES

- [1] J. E. van Zyl and C. R. I. Clayton, "The effect of pressure on leakage in water distribution systems," *Proceedings of the Institution of Civil Engineers - Water Management*, Volume 160 Issue 2, June 2007, pp. 109-114.
- [2] L. E. Ormsbee and K. E. Lansey, "Optimal Control of Water Supply Pumping Systems", *Journal of Water Resources Planning and Management*, Vol. 120, No. 2, 1994, pp. 237-252.
- [3] D. Bouman. Hydraulic design for gravity based water schemes. AquaShareware, Aqua for All, April 2014. https://aquaforall.org/wp-content/uploads/2014/12/Hydraulic-design-for-gravity-based-water-schemes_publication_2014.pdf.
- [4] H. Kudela. Hydraulic losses in pipes. https://www.academia.edu/25822288/Hydraulic_losses_in_pipes. Accessed on 12 May 2019.
- [5] D. Q. Mayne, J. B. Rawlings, C. V. Rao, P. O. M. Scokaert, Constrained model predictive control: Stability and optimality, *Automatica*, Vol. 36, Issue 6, June 2000, pp. 789-814.
- [6] G. Banjac, M. Vařak and M. Baotić, Adaptable urban water demand prediction system, *Water Science and Technology – Water Supply*. 15 (2015), 5, pp. 958-964.